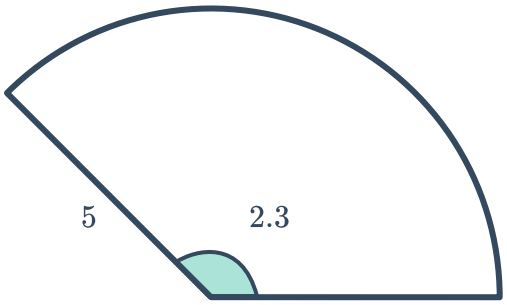
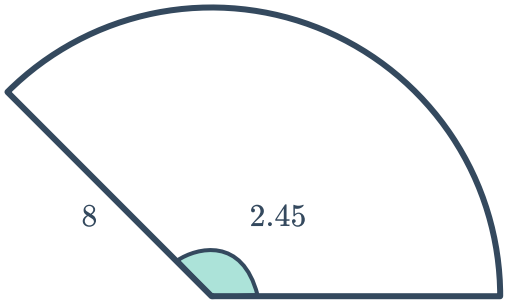
Length of circular arc

1 Find the length of the arc of the following sectors, correct to one decimal place:

a



b



2 Find the length of the arc of the following sectors, correct to two decimal places:

a A sector of a circle of radius 4 cm, formed from an angle of size $\frac{5\pi}{6}$ radians.

b A sector of a circle of diameter 19 mm, formed from an angle of size 3.5 radians.

c A sector of a circle of radius 4 cm, formed from an angle of size $\frac{3\pi}{4}$ radians.

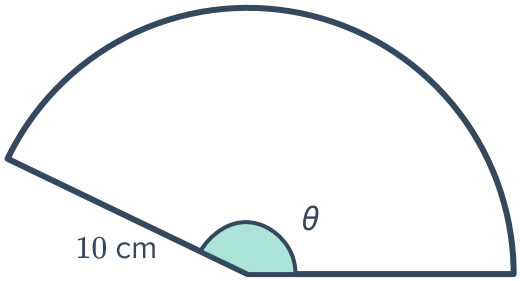
d A sector of a circle of radius 27 cm, formed from an angle of size $\frac{2}{9}$ radians.

3 The arc of a circle of radius 15 cm subtends an angle of 150° at the centre.

a Convert 150° to radians in exact form.

b Find the exact length of the arc, in terms of π.

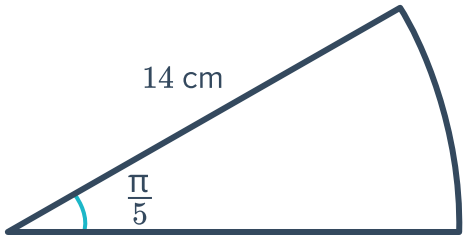
4 Consider the following sector of a circle, with radius 10 cm and central angle θ equal to $\frac{5\pi}{6}$ radians:
Find the exact value of the perimeter of the sector.



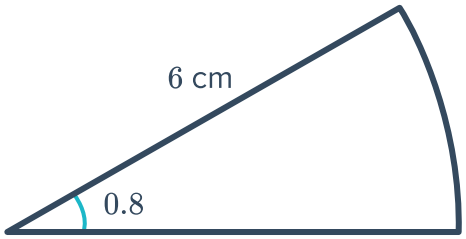
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5 Find the perimeter of the following sectors, correct to one decimal place:

a



b



6 A sector of a circle with radius $2\ \text{cm}$, has an arc length of $3.9\ \text{cm}$. Find the angle θ of the sector in radians, correct to two decimal places.

7 In each of the following, find θ, the angle subtended at the centre of the circle, in radians:

a The arc of a circle of radius $13\ \text{cm}$, measures $11.7\ \text{cm}$ in length.

b The arc of a circle of radius $15\ \text{cm}$, measures $16.5\ \text{cm}$ in length.

8 A sector of a circle with radius $95\ \text{mm}$, has an arc length of $159\ \text{mm}$.

a Find the angle θ of the sector at the centre of the circle in radians, correct to two decimal places.

b Convert the angle θ to degrees.

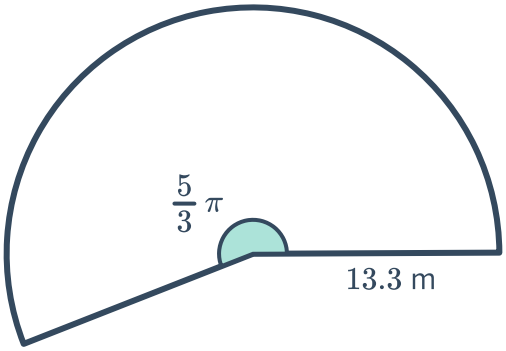
9 A sector formed from a central angle of $\frac{10}{9}\pi$ radians, is cut from a circular plastic sheet of radius $63\ \text{cm}$. The cut edges of the sector are brought together to form a plastic cone. Find the circumference of the circular base of the funnel, leaving your answer as an exact value in terms of π.

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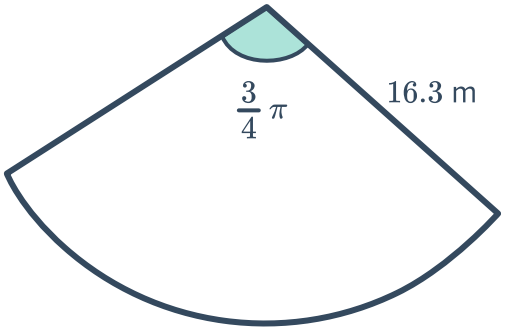
Area of a sector

10 Find the area of the following sectors, correct to two decimal places:

a



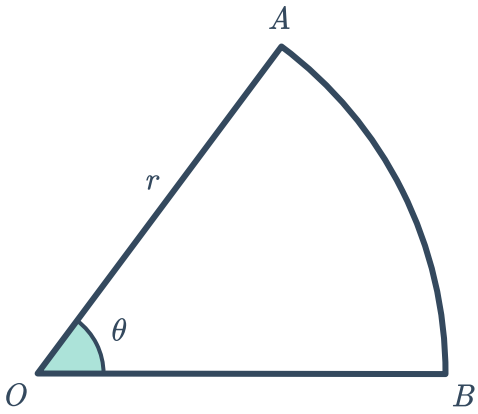
b



11 The diagram shows a sector with radius r and central angle θ:

The length of arc AB is $\frac{13\pi}{3}$ units.

Find the exact area of the sector when $r = 18$.



12 Find the area of the following sectors. Round your answer to two decimal places where necessary:

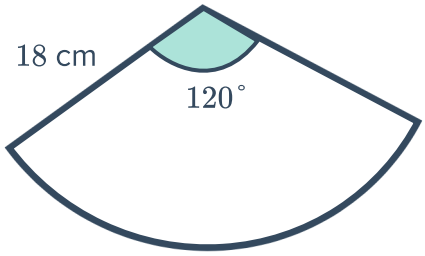
a A sector of a circle with radius 5 cm, and subtended by an angle of 4 radians at the centre.

b A sector of a circle with a radius of 24.1 m and a central angle of $\frac{4\pi}{5}$ radians.

c A sector of a circle with a radius of 22.8m and a central angle of $\frac{3\pi}{4}$ radians.

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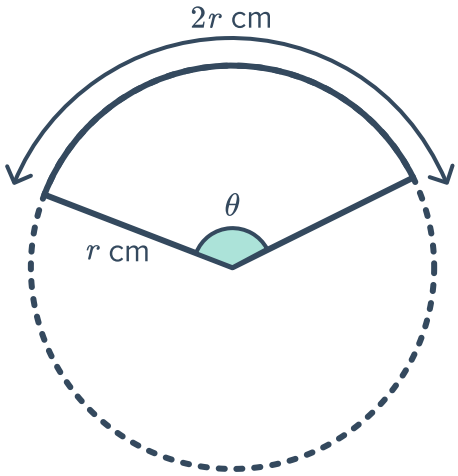
13 Consider the following sector:



a Convert the central angle of 120° to radians.

b Find the exact area of the sector in terms of π.

14 The diagram shows a sector of a circle with radius of r cm. The arc subtended by angle θ measures $2r$ cm in length.

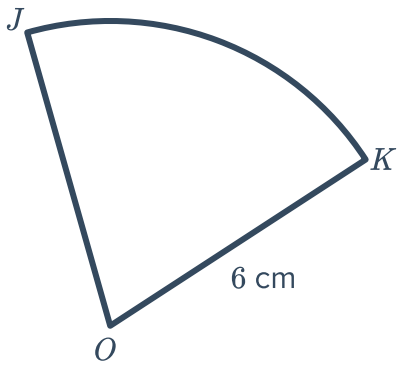


a Find central angle θ , in radians.

b Find the exact area of the sector if $r = \frac{3}{2}\pi$ cm.

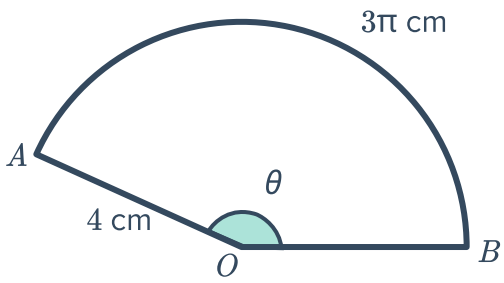
15 Find the exact area of the following sectors in terms of π:

a The following sector has radius 6 cm and arc JK has length 2π cm:

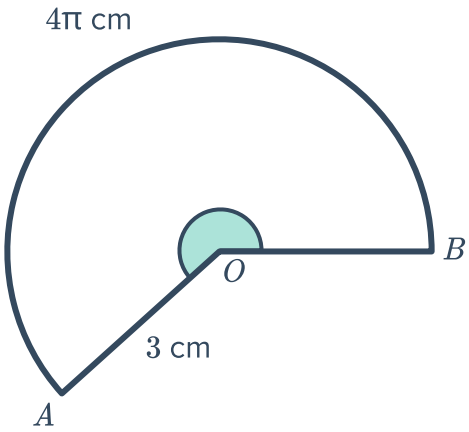


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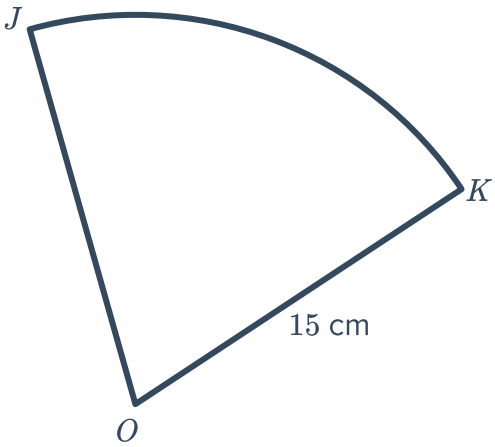
b The following sector has radius 4 cm and arc AB has length 3π cm:



c The following sector has radius 3 cm and major arc AB has length 4π cm:



16 The diagram shows an arc JK of a circle, with centre O. The radius of the circle measures 15 cm and the arc measures 12 cm in length:

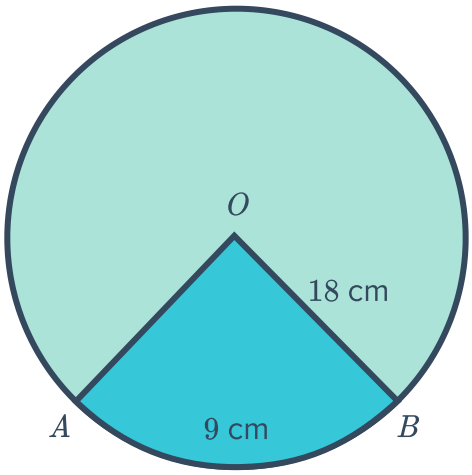


a Calculate $\angle JOK$ to the nearest degree.

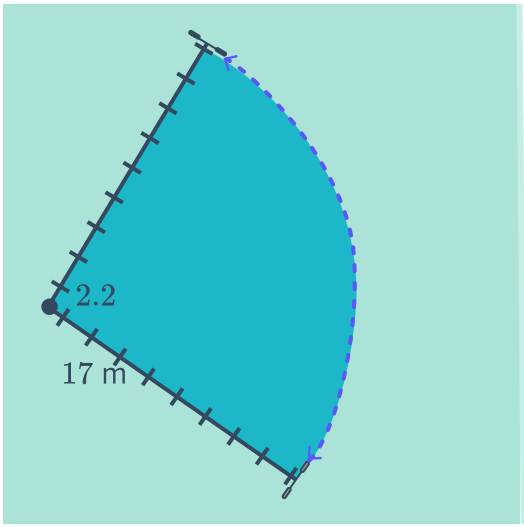
b Calculate the exact area of the sector.

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17 The given circle, centred at O, has radius 18 cm and arc AB has length 9 cm:
Find the area of the minor sector OAB.



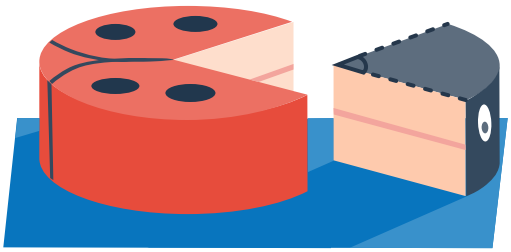
18 A large 17 m long sprinkler is placed in a crop field, with one end fixed and the other end free to move. As it turns, it waters everything underneath it.
If the sprinkler has turned an angle of 2.2 radians, find the area of the crop field it has watered.



19 Roald makes a lady beetle cake with a diameter of 24 cm and a height of 10 cm . Roald removes a slice of the cake with an angle measure of $\frac{\pi}{4}$ radians.

a Find the exact area that the black icing covers on the removed slice (both the top and side).

b Find the exact area that the red icing covers on the remaining cake (both the top and side).

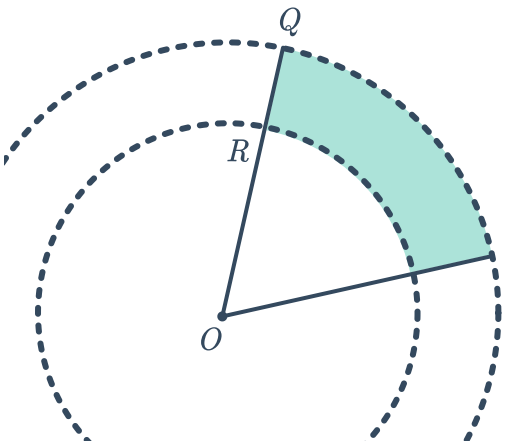


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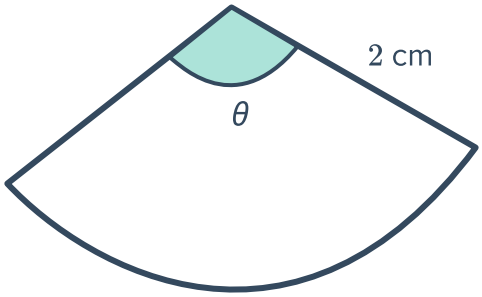
20 The diagram shows the sectors of two concentric circles with common centre O where $\angle O = \frac{\pi}{4}$ radians, $OR = 8$ cm, and $OQ = 12$ cm.

a Calculate the exact area of the shaded region between the two circular arcs.

b Calculate the exact perimeter of the shaded region.



21 The given sector has an area of 3.4 cm^2 and a radius of 2 cm:
Find the value of θ in radians.



22 A circle has a centre at O and a radius r of 20 m . Sector OAC has an included angle of θ radians, and an arc length l m. The area of sector OAC is 160 m^2.

a Find θ, the measure of the angle $\angle AOC$ in radians.

b Find the value of l.

23 The area of a sector of a circle with a radius of 9 cm is 49.5 cm^2. Find the length of the arc of the sector, l.

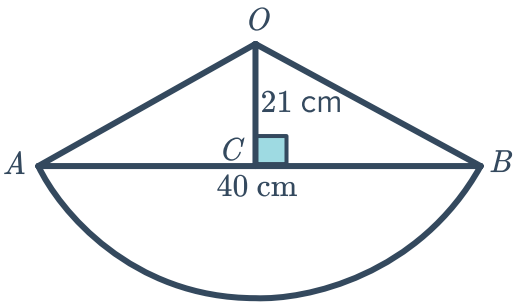
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Triangles in sectors

24 The diagram shows a chord AB which is 40 cm long, and has a perpendicular distance of 21 cm from the centre of the circle, O.

a Find the angle $\angle AOC$ in radians, correct to two decimal places.

b Calculate the length of the minor arc AB, correct to two decimal places.



25 An arc AB subtends an angle of 90° at the centre of a circle of radius $6\sqrt{2}$ cm.

a Find the length of the chord AB.

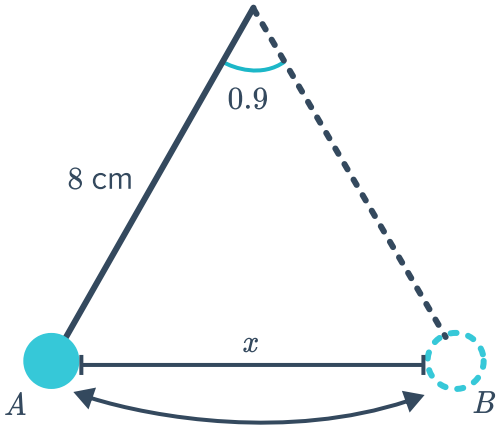
b Find the exact difference in length between the minor arc AB and the chord AB.

26 A pendulum is 8 cm long and swings through an angle of 0.9 radians. The extreme points of the pendulum are indicated by points A and B in the diagram:

a Find the length of the arc AB, correct to one decimal place.

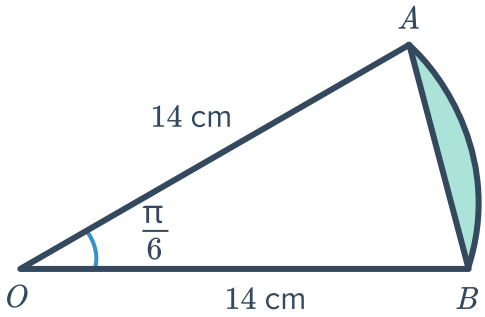
b Find x, the straight line distance between the points A and B, correct to one decimal place.

c Find the area of the sector swept out by the pendulum, correct to one decimal place.



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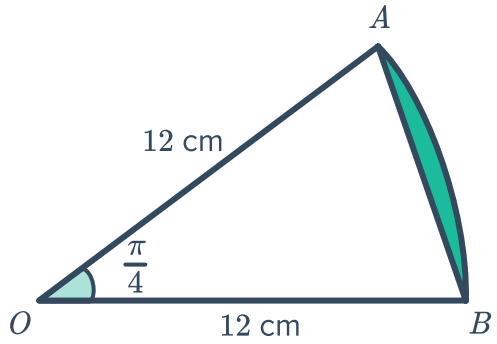
27 In the diagram, OAB is a sector of a circle with radius 14 cm, where the central angle is $\frac{\pi}{6}$ radians:



a Find the exact value of the area of the triangle OAB.

b Find the exact value of the area of the shaded segment.

28 Find the exact area of the shaded segment of the following circle:

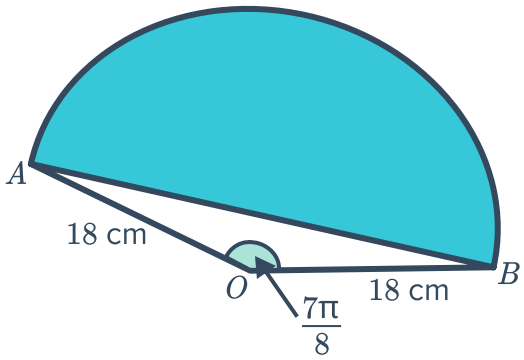


29 For each of the following sectors:

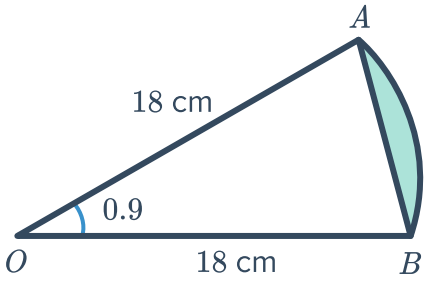
i Find the area of the triangle OAB, correct to two decimal places.

ii Find the area of the shaded segment, correct to two decimal places.

a



b

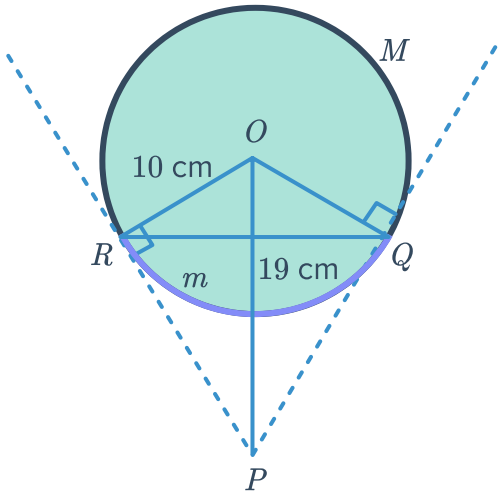


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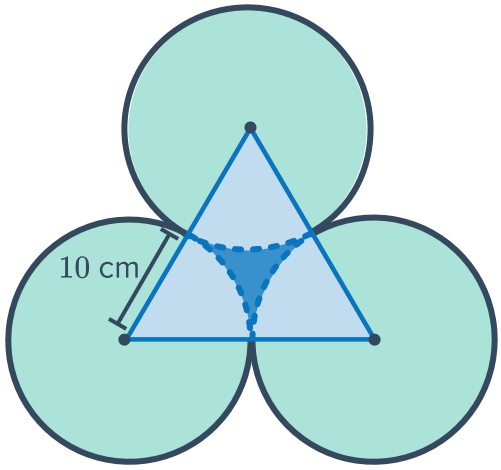
30 A point P lies $19\ \text{cm}$ from the centre of a circle with a radius of $10\ \text{cm}$. The two tangents through the point P contact the circle at R and Q:

a Find the value of $\angle ROP$ in radians, correct to two decimal places.

b Hence, find the length of the major arc M, correct to two decimal places.



31 The diagram shows three circles with radii $r = 10\ \text{cm}$. The centres form the vertices of an equilateral triangle.
Find the exact area of the space formed between the three circles.



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